

Response to the Reproduction of "From alternative conceptions of honesty to alternative facts in communications by US politicians"

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We would like to first thank the authors of the replication report, Benjamin Brummernhenrich, Daniela Feistauer, Wibke Eickmann and Elena Vaporova (BFEV from here on) for their effort. We are happy to see that they successfully reproduced one of our results related to the partisan and temporal dynamics of conceptions of honesty. However, it is unfortunate that BFEV did not attempt to replicate our main finding, namely that fact-speaking and belief-speaking are related to the trustworthiness of information that is shared by US politicians online. As far as we can understand from the replication report, BFEV only attempted a *reproduction* of our work by executing the scripts provided in our replication package. They stopped their reproduction attempts when encountering problems such as a path to a remote server that they would have needed to replace, a software version conflict, or a missing data file with intermediate results. Below, we want to comment both on the attempted *reproduction* as well as the *replicability* of our work.

We recognize two major problems that hinder *reproducibility* of our work. First, data sharing constraints due to restrictions on sharing proprietary (NewsGuard scores) or personal (tweet texts) data make the provision of all materials necessary for a direct reproduction legally impossible. Second, the complexity of the workflow involving multiple programming languages, data sets, and remote computing infrastructure to process large amounts of data makes it near impossible to create a reproduction package that executes in a different environment without modifications. To alleviate these problems, we have introduced a number of changes to the replication package that should enable researchers to directly reproduce the test of our main hypothesis and the corresponding Figure 3 from the original manuscript. Specifically, we now provide a number of data files with intermediary results of previous processing steps that are ingested by the scripts that fit the statistical models and produce visualizations. In addition, and following the suggestion of BFEV, we have added a file listing all dependencies of the code on third-party libraries that enables installing those dependencies with a single command. All changes made to the provided materials are listed in Table 1 below and are tracked in the corresponding GitHub and OSF repositories.

Given the abovementioned challenges involved in reproducing our work, we understand the materials provided by us as a *replication package* intended to provide resources and pointers that enable other researchers to achieve the same scientific findings based on the original method but with modifications, such as acquiring their own data and computing resources. For example, we point to alternative trustworthiness scores that can replace the NewsGuard data in the analysis (see <https://doi.org/10.5281/ZENODO.6536692>). In addition, we invite interested researchers to apply for sharing of the original Twitter data which we will negotiate in coordination with our institutions' research governance staff. In summary, we believe that while reproducibility without any modification to the provided materials or contact to the original authors in general is a desirable goal, it is not feasible for research like ours due to the complexity of the analyses and data sharing restrictions.

Table 1

Changes made to files and the provided materials to improve reproducibility.

Code File	Changes made
get_US_politician_twitter_accounts.ipynb	We have checked the original links to the sources of Twitter accounts of house and senate members and found that at least one is not accessible anymore. Due to this and the increased link rot currently happening on US-based government- and research websites we decided to upload the original files to the OSF repository such that reproduction of subsequent scripts does not depend on successful running of this script.
get_US_politician_twitter_timelines.ipynb	We removed any reference to our server, which was only there to serve as an example of how to run the associated code on a remote server. We still highly recommend running this script on a remote server due to the relatively long time the script takes to run (days). The script will however still not execute correctly without access to the X/Twitter API.
hydrate_tweets.ipynb	We fixed a bug that prevented the script from running successfully. However, the script still relies on access to the X/Twitter API to execute correctly.
article_scraping.ipynb	We removed the script since it should not have been part of the replication package in the first place.
article_scraping.r	We have adapted the script such that it does not need the column “score” to de-duplicate articles and can be run with the files provided in the replication package.
lmer_models_tweets.Rmd	No changes were made to the script but we now provide the following files to enable reproduction of the results in the corresponding OSF repository: • tweets_for_lme_modelling_independent.csv • tweets_for_lme_modelling_NG.csv • tweets_for_lme_modelling_other_embeddings.csv
document_level_validation.R	We changed the titles of the figures in the script to match the titles of the published figures (e.g., renaming “truth-seeking” to “fact-speaking”).
plots.ipynb	No changes were made to the script but we now provide the following files to enable reproduction of the results in the corresponding OSF repository: • LME_predictions_tweets_belief.csv • LME_predictions_tweets_truth.csv • LME_predictions_tweets_threeway.csv
requirements.txt	Added a file that records the dependencies on third party libraries needed to run the scripts. The dependencies can be installed using <code>pip install requirements.txt</code>

